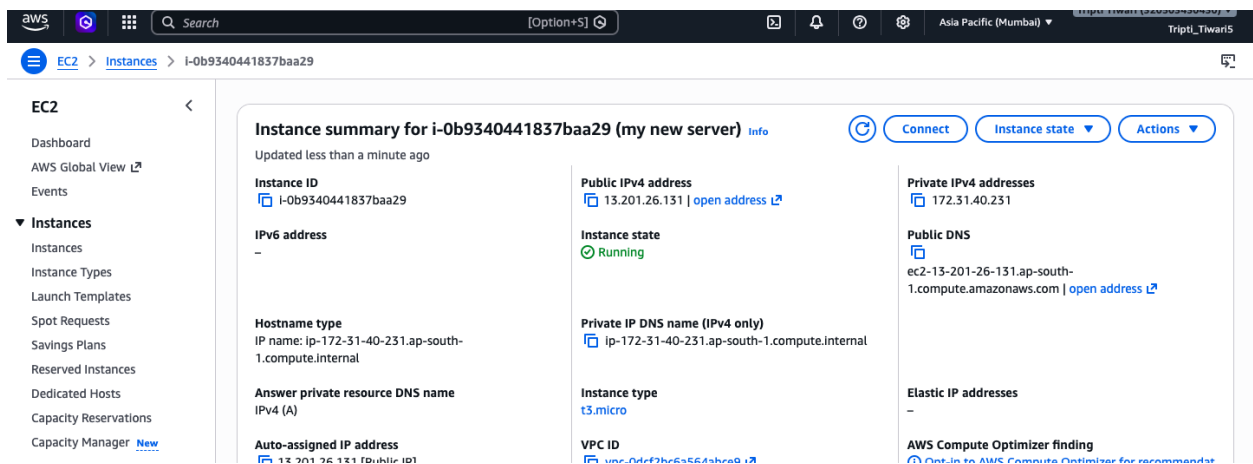


AWS 3-Tier Web Application with Docker

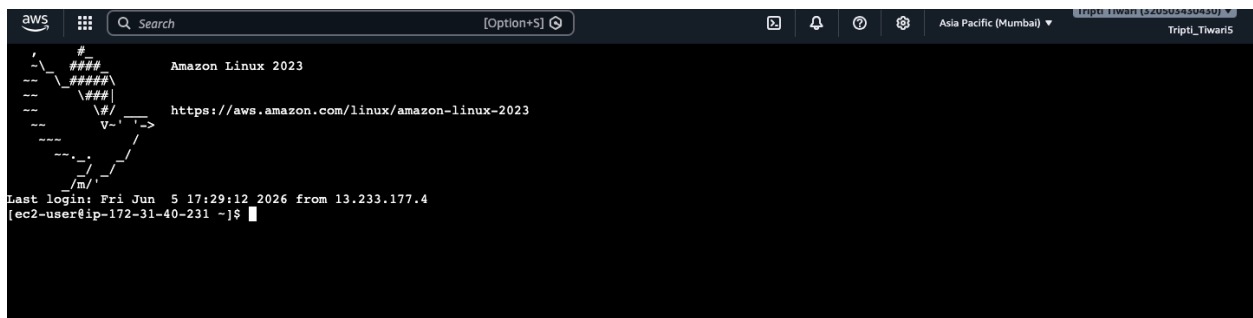
Compact Guide

STEP 1:- Launch EC2



Launch Amazon Linux EC2 and allow ports 22, 80, and 5000.

`ssh -i key.pem ec2-user@<public-ip>`



STEP 2:- Install Docker

`sudo yum update -y`

`sudo yum install docker -y`

`sudo systemctl start docker`

`sudo systemctl enable docker`

`docker --version`

```
Last login: Fri Jun  5 17:29:12 2026 from 13.233.177.4
[ec2-user@ip-172-31-40-231 ~]$ sudo yum update -y
Last metadata expiration check: 22:58:29 ago on Fri Jun  5 07:25:41 2026.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-40-231 ~]$ sudo yum install docker -y
Last metadata expiration check: 22:58:52 ago on Fri Jun  5 07:25:41 2026.
Package docker-25.0.14-1.amzn2023.0.6.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-40-231 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-40-231 ~]$ sudo systemctl enable docker
[ec2-user@ip-172-31-40-231 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-40-231 ~]$
```

i-0b9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

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STEP 3:- Project Structure

`student-app/`

`frontend/`

`backend/`

docker-compose.yml

```
Complete!
[ec2-user@ip-172-31-40-231 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-40-231 ~]$ sudo systemctl enable docker
[ec2-user@ip-172-31-40-231 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-40-231 ~]$ student-app/
-bash: student-app/: No such file or directory
[ec2-user@ip-172-31-40-231 ~]$ frontend/
-bash: frontend/: No such file or directory
[ec2-user@ip-172-31-40-231 ~]$ nano index.html
[ec2-user@ip-172-31-40-231 ~]$ backend/
-bash: backend/: No such file or directory
[ec2-user@ip-172-31-40-231 ~]$ docker-compose.yml
-bash: docker-compose.yml: command not found
[ec2-user@ip-172-31-40-231 ~]$ sudo usermod -aG docker ec2-user
[ec2-user@ip-172-31-40-231 ~]$ newgrp docker
[ec2-user@ip-172-31-40-231 ~]$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
01a14a57af52   student-portfolio  "/docker-entrypoint..." 23 hours ago    Up 23 hours    0.0.0.0:80->80/tcp, :::80->80/tcp  inspiring_robinson
[ec2-user@ip-172-31-40-231 ~]$ mkdir student-app
```

STEP 4:- Step 6: Create Project Structure

mkdir student-app

cd student-app

Create folders:

mkdir frontend

mkdir backend

Check:

ls

```

-bash: frontend/: No such file or directory
[ec2-user@ip-172-31-40-231 ~]$ nano index.html
[ec2-user@ip-172-31-40-231 ~]$ backend/
-bash: backend/: No such file or directory
[ec2-user@ip-172-31-40-231 ~]$ docker-compose.yml
-bash: docker-compose.yml: command not found
[ec2-user@ip-172-31-40-231 ~]$ sudo usermod -aG docker ec2-user
[ec2-user@ip-172-31-40-231 ~]$ newgrp docker
[ec2-user@ip-172-31-40-231 ~]$ docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS                    NAMES
01a14a57af52   student-portfolio   "/docker-entrypoint..." 23 hours ago   Up 23 hours   0.0.0.0:80->80/tcp, :::80->80/tcp   inspiring_robinson
[ec2-user@ip-172-31-40-231 ~]$ mkdir student-app
[ec2-user@ip-172-31-40-231 ~]$ cd student-app
[ec2-user@ip-172-31-40-231 student-app]$ mkdir frontend
[ec2-user@ip-172-31-40-231 student-app]$ mkdir backend
[ec2-user@ip-172-31-40-231 student-app]$ ls
backend  frontend
[ec2-user@ip-172-31-40-231 student-app]$ cd frontend
[ec2-user@ip-172-31-40-231 frontend]$ nano index.html

```

i-0b9340441837baa29 (my new server)

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Step 5: Create Frontend

Go inside:

cd frontend

Create file:

nano index.html

Paste:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Student Registration</title>
```

```
</head>
```

```
<body>
```

```
<h1>Student Registration</h1>
```

```
<form>
```

```
<input placeholder="Name"><br><br>
<input placeholder="Email"><br><br>
<input placeholder="Course"><br><br>

<button>Submit</button>
</form>
```

```
</body>
</html>
```

Save:

CTRL + O

ENTER

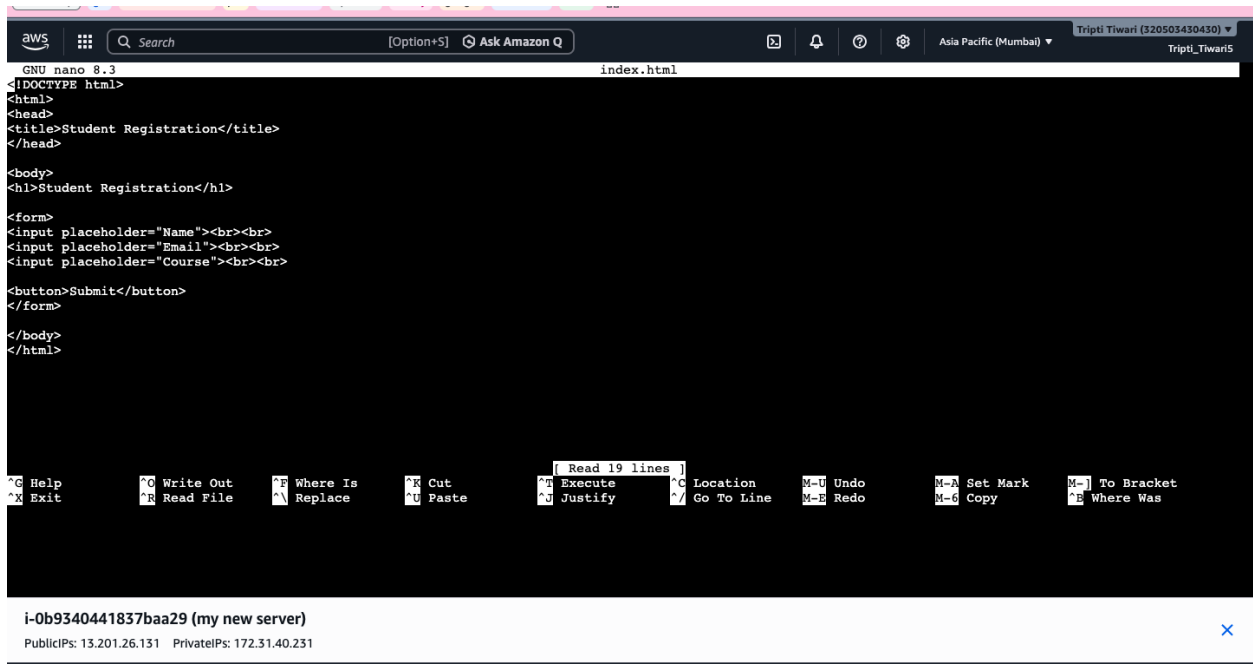
CTRL + X

```
[ec2-user@ip-172-31-40-231 student-app]$ mkdir backend
[ec2-user@ip-172-31-40-231 student-app]$ ls
backend  frontend
[ec2-user@ip-172-31-40-231 student-app]$ cd frontend
[ec2-user@ip-172-31-40-231 frontend]$ nano index.html
```

i-0b9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231





```
GNU nano 8.3 index.html
<!DOCTYPE html>
<html>
<head>
<title>Student Registration</title>
</head>

<body>
<h1>Student Registration</h1>

<form>
<input placeholder="Name"><br><br>
<input placeholder="Email"><br><br>
<input placeholder="Course"><br><br>

<button>Submit</button>
</form>

</body>
</html>
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location ^U Undo ^M Set Mark ^= To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line ^_ Redo ^- Copy ^_ Where Was

i-0b9340441837baa29 (my new server)
PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

Save:

Ctrl +S

CTRL + O

ENTER

CTRL + X

Step 6: Frontend Dockerfile

Create:

nano Dockerfile

Paste:

FROM nginx

COPY ./usr/share/nginx/html

Save.

```
(ec2-user@ip-172-31-40-231 ~)$ cd student-app
(ec2-user@ip-172-31-40-231 student-app)$ mkdir frontend
(ec2-user@ip-172-31-40-231 student-app)$ mkdir backend
(ec2-user@ip-172-31-40-231 student-app)$ ls
backend frontend
(ec2-user@ip-172-31-40-231 student-app)$ cd frontend
(ec2-user@ip-172-31-40-231 frontend)$ nano index.html
(ec2-user@ip-172-31-40-231 frontend)$ nano Dockerfile
(ec2-user@ip-172-31-40-231 frontend)$ cd ..
(ec2-user@ip-172-31-40-231 student-app)$ cd backend
(ec2-user@ip-172-31-40-231 backend)$ nano app.py
(ec2-user@ip-172-31-40-231 backend)$ nano requirements.txt
(ec2-user@ip-172-31-40-231 backend)$
```

i-0b9340441837baa29 (my new server)

The screenshot shows an AWS CloudShell terminal window. The top bar includes the AWS logo, a search bar, and the region 'Asia Pacific (Mumbai)'. The terminal output shows the following commands and their results:

```
GNU nano 8.3 Dockerfile
FROM nginx
COPY ./usr/share/nginx/html
```

The bottom of the terminal window shows a list of keyboard shortcuts for nano, such as 'H Help', 'X Exit', 'W Write Out', 'R Read File', etc. The terminal window is titled 'i-0b9340441837baa29 (my new server)' and shows the public and private IP addresses.

Step 9: Create Backend

Go back:

```
cd ..
```

```
cd backend
```

Create app:

```
nano app.py
```

Paste:

```
from flask import Flask
```

```
app = Flask(__name__)
```

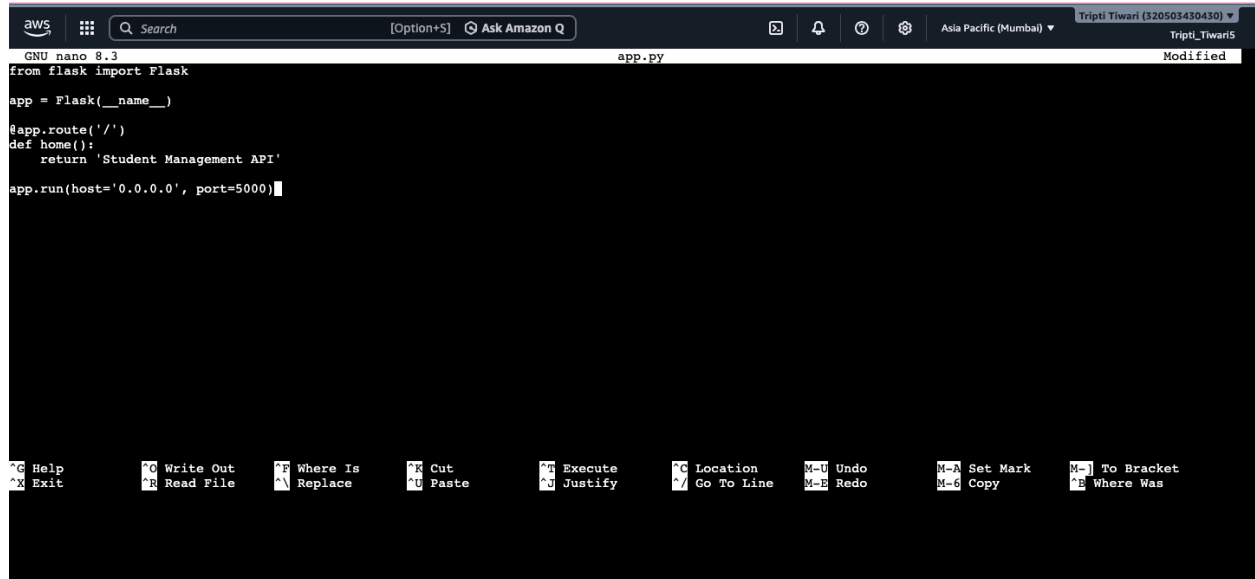
```
@app.route('/')
```

```
def home():
```

```
    return 'Student Management API'
```

```
app.run(host='0.0.0.0', port=5000)
```

Save.



```
GNU nano 8.3 app.py
from flask import Flask

app = Flask(__name__)

@app.route('/')
def home():
    return 'Student Management API'

app.run(host='0.0.0.0', port=5000)
```

^G Help ^O Write Out ^F Where Is ^K Cut ^I Execute ^C Location ^U Undo ^M-A Set Mark ^M-J To Bracket
^X Exit ^R Read File ^A Replace ^U Paste ^J Justify ^_ Go To Line ^M-U Redo ^M-C Copy ^M-W Where Was

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Step 10: Requirements File

Create:

nano requirements.txt

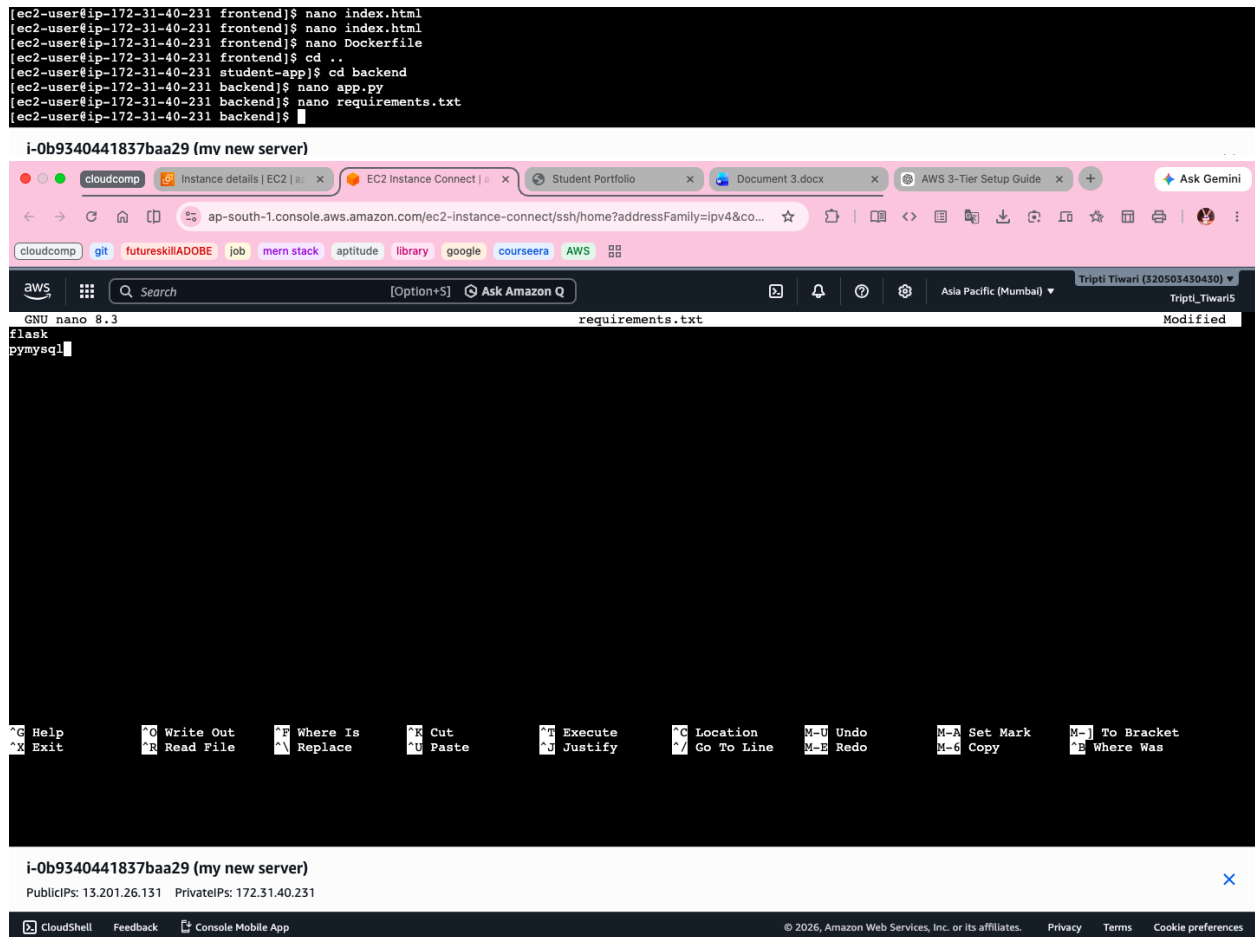
Paste:

flask

pymysql

Save.

```
[ec2-user@ip-172-31-40-231 frontend]$ nano index.html
[ec2-user@ip-172-31-40-231 frontend]$ nano index.html
[ec2-user@ip-172-31-40-231 frontend]$ nano Dockerfile
[ec2-user@ip-172-31-40-231 frontend]$ cd ..
[ec2-user@ip-172-31-40-231 student-app]$ cd backend
[ec2-user@ip-172-31-40-231 backend]$ nano app.py
[ec2-user@ip-172-31-40-231 backend]$ nano requirements.txt
[ec2-user@ip-172-31-40-231 backend]$
```



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Step 11: Backend Dockerfile

Create:

nano Dockerfile

Paste:

FROM python:3.11

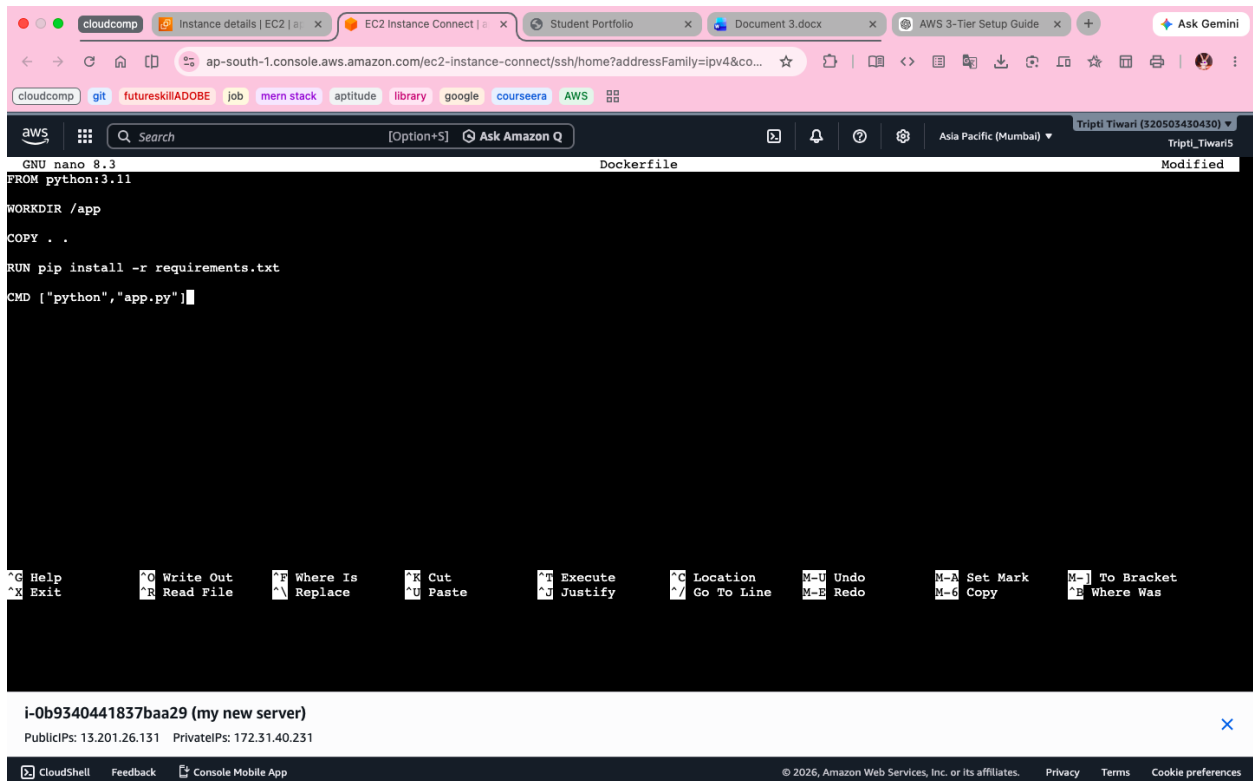
WORKDIR /app

COPY ..

RUN pip install -r requirements.txt

CMD ["python","app.py"]

Save.



```
GNU nano 8.3 Dockerfile
FROM python:3.11

WORKDIR /app

COPY . .

RUN pip install -r requirements.txt

CMD ["python", "app.py"]
```

i-0b9340441837baa29 (my new server)
PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

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Step 12: Create docker-compose.yml

Go to root folder:

cd ..

Create:

nano docker-compose.yml

Paste:

version: '3'

services:

frontend:

build: ./frontend

ports:

- "80:80"

backend:

build: ./backend

ports:

- "5000:5000"

Save.

The screenshot shows the AWS CloudShell interface. At the top, there are browser tabs for 'cloudcomp', 'Instance details | EC2 | s...', 'EC2 Instance Connect | s...', 'Student Portfolio', 'Document 3.docx', 'AWS 3-Tier Setup Guide', and 'Ask Gemini'. The address bar shows 'ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&co...'. Below the browser tabs is a navigation bar with 'cloudcomp', 'git', 'futureskill', 'ADOBE', 'job', 'mern stack', 'aptitude', 'library', 'google', 'courseera', 'AWS', and a search bar. The main terminal area shows the nano 8.3 editor editing 'docker-compose.yml'. The file content is:

```
version: '3'

services:
  frontend:
    build: ./frontend
    ports:
      - "80:80"
  backend:
    build: ./backend
    ports:
      - "5000:5000"
```

At the bottom of the terminal, there is a status bar with 'i-0b9340441837baa29 (my new server)', 'PublicIPs: 13.201.26.131', and 'PrivateIPs: 172.31.40.231'. Below the terminal is a footer with 'CloudShell', 'Feedback', 'Console Mobile App', and copyright information.

Step 13: Build Containers

Run:

`docker compose up -d`

If it says command not found:

`sudo yum install docker-compose-plugin -y`

Then:

`docker compose up -d`

cloudcomp Instance details | EC2 | x EC2 Instance Connect | x Student Portfolio x Document 3.docx x AWS 3-Tier Setup Guide x + Ask Gemini

ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&co...

cloudcomp git futureskillADOBE job mern stack aptitude library google courseera AWS

aws Search [Option+S] Ask Amazon Q

Asia Pacific (Mumbai) Tripti Tiwari (320503430430) Tripti_Tiwari5

logs Fetch the logs of a container
pause Pause all processes within one or more containers
port List port mappings or a specific mapping for the container
rename Rename a container
restart Restart one or more containers
rm Remove one or more containers
rmi Remove one or more images
save Save one or more images to a tar archive (streamed to STDOUT by default)
start Start one or more stopped containers
stats Display a live stream of container(s) resource usage statistics
stop Stop one or more running containers
tag Create a tag TARGET_IMAGES that refers to SOURCE_IMAGE
top Display the running processes of a container
unpause Unpause all processes within one or more containers
update Update configuration of one or more containers
wait Block until one or more containers stop, then print their exit codes

Global Options:
--config string Location of client config files (default "/home/ec2-user/.docker")
-c, --context string Name of the context to use to connect to the daemon (overrides DOCKER_HOST env var and default context set with "docker context use")
-D, --debug Enable debug mode
-H, --host list Daemon socket to connect to
-l, --log-level string Set the logging level ("debug", "info", "warn", "error", "fatal") (default "info")
--tls Use TLS; implied by --tlsverify
--tlscacert string Trust certs signed only by this CA (default "/home/ec2-user/.docker/ca.pem")
--tlscert string Path to TLS certificate file (default "/home/ec2-user/.docker/cert.pem")
--tlskey string Path to TLS key file (default "/home/ec2-user/.docker/key.pem")
--tlsverify Use TLS and verify the remote
-v, --version Print version information and quit

Run 'docker COMMAND --help' for more information on a command.

For more help on how to use Docker, head to <https://docs.docker.com/go/guides/>

[ec2-user@ip-172-31-40-231 student-app]\$

i-0b9340441837baa29 (my new server)

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cloudcomp Instance details | EC2 | x EC2 Instance Connect | x Student Portfolio x Document 3.docx x AWS 3-Tier Setup Guide x + Ask Gemini

ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&co...

cloudcomp git futureskillADOBE job mern stack aptitude library google courseera AWS

aws Search [Option+S] Ask Amazon Q

Asia Pacific (Mumbai) Tripti Tiwari (320503430430) Tripti_Tiwari5

[ec2-user@ip-172-31-40-231 ~]\$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-40-231 ~]\$ student-app/
-bash: student-app/: No such file or directory
[ec2-user@ip-172-31-40-231 ~]\$ frontend/
-bash: frontend/: No such file or directory
[ec2-user@ip-172-31-40-231 ~]\$ nano index.html
[ec2-user@ip-172-31-40-231 ~]\$ backend/
-bash: backend/: No such file or directory
[ec2-user@ip-172-31-40-231 ~]\$ docker-compose.yml
-bash: docker-compose.yml: command not found
[ec2-user@ip-172-31-40-231 ~]\$ sudo usermod -sG docker ec2-user
[ec2-user@ip-172-31-40-231 ~]\$ newgrp docker
[ec2-user@ip-172-31-40-231 ~]\$ docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
01a14a57af52	student-portfolio	"/docker-entrypoint..."	23 hours ago	Up 23 hours	0.0.0.0:80->80/tcp, :::80->80/tcp	inspiring_robinson

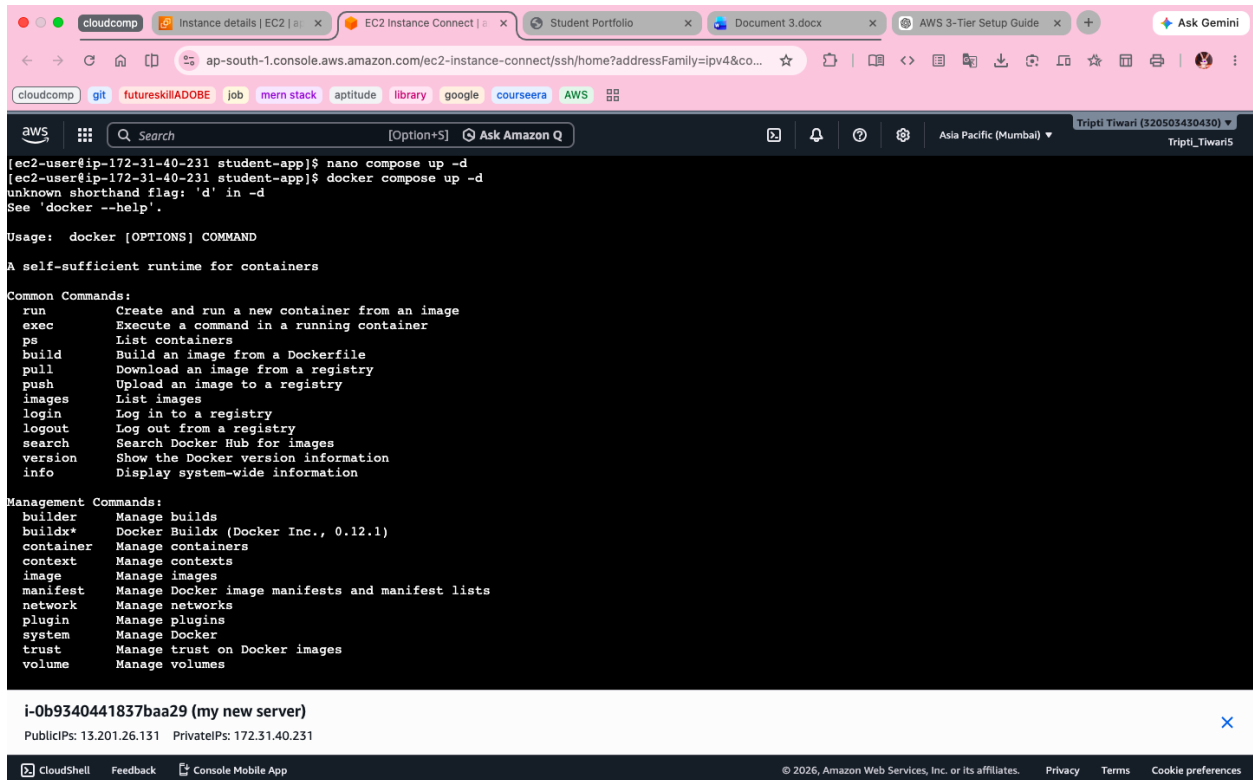
[ec2-user@ip-172-31-40-231 ~]\$ mkdir student-app
[ec2-user@ip-172-31-40-231 ~]\$ cd student-app
[ec2-user@ip-172-31-40-231 student-app]\$ mkdir frontend
[ec2-user@ip-172-31-40-231 student-app]\$ mkdir backend
mkdir: cannot create directory 'frontend': File exists
[ec2-user@ip-172-31-40-231 student-app]\$ mkdir backend
[ec2-user@ip-172-31-40-231 student-app]\$ ls
backend frontend
[ec2-user@ip-172-31-40-231 student-app]\$ cd frontend
[ec2-user@ip-172-31-40-231 frontend]\$ nano index.html
[ec2-user@ip-172-31-40-231 frontend]\$ nano Dockerfile
[ec2-user@ip-172-31-40-231 frontend]\$ cd ..
[ec2-user@ip-172-31-40-231 student-app]\$ cd backend
[ec2-user@ip-172-31-40-231 backend]\$ nano app.py
[ec2-user@ip-172-31-40-231 backend]\$ nano requirements.txt
[ec2-user@ip-172-31-40-231 backend]\$ nano Dockerfile
[ec2-user@ip-172-31-40-231 backend]\$ cd ..
[ec2-user@ip-172-31-40-231 student-app]\$ nano docker-compose.yml
[ec2-user@ip-172-31-40-231 student-app]\$ nano compose up -d

i-0b9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

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The screenshot shows an AWS CloudShell terminal window. The top navigation bar includes tabs for 'cloudcomp', 'Instance details | EC2 | s...', 'EC2 Instance Connect | s...', 'Student Portfolio', 'Document 3.docx', 'AWS 3-Tier Setup Guide', and 'Ask Gemini'. The address bar shows the URL 'ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&co...'. Below the navigation bar, there are links for 'cloudcomp', 'git', 'futureskill', 'ADOBE', 'job', 'mern stack', 'aptitude', 'library', 'google', 'courseera', and 'AWS'. The terminal window displays the following text:

```
[ec2-user@ip-172-31-40-231 student-app]$ nano compose up -d
[ec2-user@ip-172-31-40-231 student-app]$ docker compose up -d
unknown shorthand flag: 'd' in -d
See 'docker --help'.
```

Usage: docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:

run	Create and run a new container from an image
exec	Execute a command in a running container
ps	List containers
build	Build an image from a Dockerfile
pull	Download an image from a registry
push	Upload an image to a registry
images	List images
login	Log in to a registry
logout	Log out from a registry
search	Search Docker Hub for images
version	Show the Docker version information
info	Display system-wide information

Management Commands:

builder	Manage builds
buildx	Docker Buildx (Docker Inc., 0.12.1)
container	Manage containers
context	Manage contexts
image	Manage images
manifest	Manage Docker image manifests and manifest lists
network	Manage networks
plugin	Manage plugins
system	Manage Docker
trust	Manage trust on Docker images
volume	Manage volumes

i-0b9340441837baa29 (my new server)

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Step 14: Verify Containers

docker ps

The screenshot shows an AWS CloudShell terminal window. The top navigation bar includes tabs for 'cloudcomp', 'Instance details | EC2 |', 'EC2 Instance Connect |', 'Student Portfolio', 'Document 3.docx', and 'AWS 3-Tier Setup Guide'. The address bar shows the URL 'ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&co...'. The terminal window has a search bar and a list of Docker commands with their descriptions. Below this, the 'Global Options' are listed. The output of the 'docker ps' command is shown as a table. At the bottom, there is a notification for a new server and a footer with 'CloudShell', 'Feedback', 'Console Mobile App', and copyright information.

```
rename      Rename a container
restart     Restart one or more containers
rm          Remove one or more containers
rmi         Remove one or more images
save        Save one or more images to a tar archive (streamed to STDOUT by default)
start       Start one or more stopped containers
stats       Display a live stream of container(s) resource usage statistics
stop        Stop one or more running containers
tag         Create a tag TARGET_IMAGE that refers to SOURCE_IMAGE
top         Display the running processes of a container
unpause     Unpause all processes within one or more containers
update      Update configuration of one or more containers
wait        Block until one or more containers stop, then print their exit codes

Global Options:
--config string      Location of client config files (default "/home/ec2-user/.docker")
-c, --context string  Name of the context to use to connect to the daemon (overrides DOCKER_HOST env var and default context set with "docker context use")
-D, --debug           Enable debug mode
-H, --host list       Daemon socket to connect to
-l, --log-level string Set the logging level ("debug", "info", "warn", "error", "fatal") (default "info")
--tls                Use TLS; implied by --tlsverify
--tlscacert string    Trust certs signed only by this CA (default "/home/ec2-user/.docker/ca.pem")
--tlscert string      Path to TLS certificate file (default "/home/ec2-user/.docker/cert.pem")
--tlskey string       Path to TLS key file (default "/home/ec2-user/.docker/key.pem")
--tlsverify           Use TLS and verify the remote
-v, --version         Print version information and quit

Run 'docker COMMAND --help' for more information on a command.

For more help on how to use Docker, head to https://docs.docker.com/go/guides/

[ec2-user@ip-172-31-40-231 student-app]$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
01a14a57af52   student-portfolio  "/docker-entrypoint..." 23 hours ago Up 23 hours  0.0.0.0:80->80/tcp, :::80->80/tcp  inspiring_robinson
[ec2-user@ip-172-31-40-231 student-app]$
```

i-0b9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

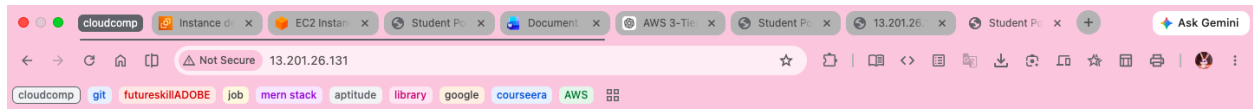
CloudShell Feedback Console Mobile App

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Step 15: Open in Browser

Frontend:

http://YOUR_PUBLIC_IP



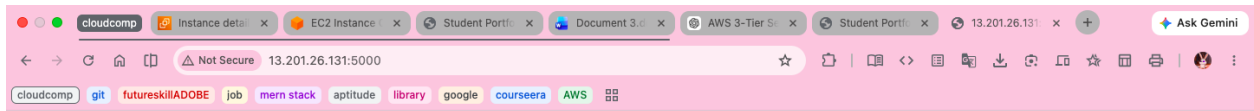
Welcome to My Portfolio

Name: Student Name

Skills: AWS, Python, Docker

Backend:

http://YOUR_PUBLIC_IP:5000



Student Management API